

Pulkit Tiwari

tpulkit87@gmail.com | +91 7014167303 | linkedin.com/in/pulkittiwari51 | github.com/PulkitTiwari87 | Portfolio

Summary

Computer Science undergraduate specializing in Cybersecurity and Forensics, with experience building Generative AI systems, distributed backend platforms, security automation, and full-stack applications through internships, freelance work, and projects. Strong foundation in data structures, algorithms, machine learning, and LLM applications.

Education

University of Petroleum and Energy Studies (UPES), Dehradun
B.Tech Computer Science & Engineering – Cybersecurity and Forensics

Aug 2023 – May 2027
CGPA: 7.43/10

Skills

Languages & Core CS: Python, Java, C, JavaScript, TypeScript, SQL; Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Computer Networks, Distributed Systems

AI/ML/GenAI: LLMs, RAG, Agentic AI, Prompt Engineering, Embeddings, NLP, XGBoost, Scikit-learn, TensorFlow, CrewAI, AWS Bedrock, AI Guardrails, Qdrant, FAISS

Backend & Data: Node.js, Express.js, REST APIs, gRPC, Microservices, JWT, OAuth, PostgreSQL, MongoDB, MySQL, Redis

Frontend & Cloud/DevOps: React.js, Next.js, Tailwind CSS, Vite; AWS, Docker, Git, GitHub Actions, Vercel, CI/CD

Cybersecurity: Wazuh, TheHive, Cortex, MISP, ELK Stack, Kali Linux, Burp Suite, Nmap, OWASP

Experience

Software Engineering Intern, Ascendion, Bengaluru Jun 2026 – Sep 2026

- Engineered a 3-agent CrewAI pipeline on AWS Bedrock using Claude models to automate resume reformatting through structured extraction, rendering, and Azure Blob Storage upload with SAS-token authentication.
- Developed an output-validation agent using bias guardrails and knowledge-base-grounded scoring to automate compliance checks across 10+ Generative AI applications and generate standardized branded reports.

Freelance Software Developer, Upwork

Mar 2026 – May 2026

- Delivered end-to-end AI-powered SaaS and automation solutions with React.js, Node.js, Python, PostgreSQL, and LLM APIs, including MomPlan (an LLM-based benefits-eligibility platform) and an AI receptionist prototype for automated lead handling.

Full Stack Developer Intern, Weblicious, Dehradun

Jun 2025 – Jul 2025

- Architected client-facing web applications using React.js, Node.js, and MongoDB with 10+ schemas and an API-first architecture.
- Optimized REST APIs and backend workflows, improving data accessibility and reducing API response time by 25%.

Projects

SOAR Intelligence – AI-Assisted Security Operations Platform

- Built an event-driven ML and microservices security platform on Wazuh, Cortex, TheHive, MISP, ELK, Redis, and Docker for automated detection, alert triage, orchestration, and response.
- Trained an XGBoost alert-triage model on 45,000 records (15,000 synthetic, 30,000 KDD Cup '99) reaching 92.41% accuracy, 0.98 precision, and 0.92 recall; added phishing, anomaly, log-clustering, and blast-radius modules.

Atlas – Fault-Tolerant Distributed Compute Platform

- Architected a Python job-execution platform with a priority scheduler, concurrent workers, and PostgreSQL-backed job state; designed REST/gRPC APIs for scheduling, execution, and monitoring.
- Implemented and benchmarked 4 fault-tolerance mechanisms: heartbeats, failure detection, automatic retries, and graceful recovery.

Real-Time CRAG – Corrective Retrieval-Augmented Generation

- Designed a provider-agnostic Corrective RAG system combining retrieval, relevance evaluation, query refinement, web search, and grounded generation with Qdrant, FAISS, embeddings, and Hugging Face inference.

G-DriveX – Multi-Account Cloud Storage SaaS (live: g-drive-x.vercel.app)

- Built a React/Node.js SaaS unifying multiple Google Drive accounts with server-side Drive-to-Drive streaming and bulk transfers; encrypted OAuth refresh tokens at rest.
- Found and fixed IDOR and OAuth CSRF vulnerabilities; shipped with 24/24 backend tests passing and CI/CD validation.

NEO-Hazard-AI – NASA Near-Earth Object Hazard Classification (live: neo-hazard-ai.vercel.app)

- Leakage-aware ML pipeline on real NASA/JPL CNEOS data (Logistic Regression, Random Forest, XGBoost) with 5-fold CV, holdout evaluation, calibration, SHAP explainability, and anomaly detection.

Positions of Responsibility & Certifications

Public Relations & Sponsorship Head – UPES CSA Student Chapter (Cloud Security Alliance)

Sep 2024 – Apr 2026

- Led a team of 30+ organizing university technical events including hackathons, WebGenesis, and Funtopia 5.0; directed sponsorship outreach for AWS Community Day Dehradun 2025 (400+ attendees), securing industry collaboration including GitHub.

Certifications: Google Cybersecurity (Foundations of Cybersecurity; Play It Safe; Connect and Protect; Tools of the Trade: Linux and SQL); HackerRank – JavaScript (Intermediate), React, Python, CSS.